

TURKU ARTUKAINEN, Finland

WMO: 027730

Lat: 60.4544N

Lon: 22.1786E

Elev: 9

StdP: 101.22

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-15.8	-21.2	0.6	-18.2	-18.0	0.8	-15.2	9.3	-2.8	8.3	-2.0	2.0	30	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	27.6	18.8	25.6	18.2	23.7	17.2	20.6	25.2	19.4	23.8	18.2	22.4	3.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	22.7	17.8	12.8	21.4	16.6	11.8	20.0	59.3	25.1	55.2	23.6	51.4	22.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.4	5.8	-20.6	29.7	4.2	2.5	-23.7	31.5	-26.2	33.0	-28.6	34.4	-31.6	36.2		
(5)			-20.9	21.2	4.2	1.4	-23.9	22.2	-26.4	23.1	-28.7	23.9	-31.8	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-4.7	-3.3	-0.6	4.7	11.1	14.8	18.3	16.8	12.3	6.1	2.9	-0.8	(6)
(7)		DBStd	8.70	5.67	5.45	4.00	3.15	4.24	3.25	3.04	2.74	3.08	3.53	3.99	5.02	(7)
(8)		HDD10.0	2048	456	373	329	161	36	2	0	0	14	127	214	335	(8)
(9)		HDD18.3	4379	714	607	588	408	228	114	38	63	181	380	464	593	(9)
(10)		CDD10.0	776	0	0	0	3	70	147	257	210	83	5	0	0	(10)
(11)		CDD18.3	65	0	0	0	0	3	9	37	16	1	0	0	0	(11)
(12)		CDH23.3	489	0	0	0	0	36	82	293	78	1	0	0	0	(12)
(13)		CDH26.7	99	0	0	0	0	3	12	69	16	0	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.7	2.9	3.0	2.8	2.8	3.0	2.8	2.6	2.8	3.0	3.0	(14)	
(15)	Precipitation	PrecAvg	723	60	44	42	39	36	50	73	82	69	72	79	70	(15)
(16)		PrecMax	964	108	106	103	102	89	105	146	190	163	173	163	182	(16)
(17)		PrecMin	531	9	3	4	2	4	7	0	0	12	21	12	7	(17)
(18)		PrecStd	96	27	25	22	22	23	27	41	44	36	32	38	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	6.1	10.3	19.2	26.9	28.4	31.2	29.1	22.3	14.6	10.4	8.5	(19)
(20)			MCWB	4.5	3.5	5.4	10.9	17.0	18.4	20.2	20.0	17.0	11.9	9.6	7.5	(20)
(21)		2%	DB	4.3	4.4	7.7	15.6	23.9	25.7	28.6	25.3	19.8	13.1	9.4	6.9	(21)
(22)			MCWB	3.7	3.1	4.2	8.8	15.1	17.2	19.4	18.8	15.6	11.5	8.7	5.8	(22)
(23)		5%	DB	3.2	3.5	5.9	13.2	21.8	23.3	26.5	23.3	18.4	12.2	8.5	5.8	(23)
(24)			MCWB	2.5	2.6	3.3	7.6	14.3	15.7	19.1	17.5	14.7	10.8	7.7	4.9	(24)
(25)		10%	DB	2.2	2.6	4.4	10.9	19.4	21.3	24.6	21.9	17.2	11.3	7.7	5.0	(25)
(26)			MCWB	1.6	2.0	2.5	6.2	13.0	14.7	18.3	16.7	14.3	10.1	6.9	4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	4.2	6.2	11.4	18.0	20.0	22.4	21.5	17.2	12.7	10.0	7.8	(27)
(28)			MCDWB	5.0	5.1	8.8	18.3	23.5	26.6	27.5	25.5	20.2	13.6	10.2	8.3	(28)
(29)		2%	WB	3.8	3.4	4.7	9.5	16.6	18.1	21.2	19.8	16.4	12.0	8.8	6.1	(29)
(30)			MCDWB	4.2	4.1	6.6	14.6	22.4	23.8	26.1	23.7	18.9	12.8	9.2	6.6	(30)
(31)		5%	WB	2.7	2.7	3.7	8.0	15.0	16.7	20.1	18.6	15.6	11.2	7.9	5.1	(31)
(32)			MCDWB	3.1	3.4	5.3	12.2	21.1	21.9	24.8	22.0	17.8	12.0	8.5	5.7	(32)
(33)		10%	WB	1.7	2.1	2.8	6.8	13.3	15.6	19.0	17.6	14.8	10.3	7.0	4.3	(33)
(34)			MCDWB	2.2	2.7	4.1	10.2	18.9	20.1	23.5	20.8	16.6	11.1	7.6	4.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	5.0	6.6	8.7	9.8	8.6	8.7	8.1	7.2	5.8	3.9	4.4	(35)
(36)			MCDBR	3.3	4.9	7.8	13.3	13.6	12.3	11.5	10.2	8.8	6.0	4.1	4.7	(36)
(37)			MCWBR	3.2	4.1	5.3	7.5	6.5	5.5	4.5	4.6	5.1	4.6	3.8	4.3	(37)
(38)		5% WB	MCDBR	3.3	4.6	6.9	11.6	12.7	10.6	9.7	8.3	7.3	5.0	3.9	4.3	(38)
(39)			MCWBR	3.3	4.0	5.1	6.9	6.3	5.4	4.3	4.3	4.7	4.4	3.8	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.314	0.334	0.373	0.369	0.361	0.389	0.379	0.350	0.333	0.302	0.384	(40)	
(41)		taud	2.151	2.234	2.309	2.317	2.400	2.451	2.394	2.416	2.484	2.395	2.210	2.602	(41)	
(42)		Ebn at Noon	468	668	789	822	857	870	831	810	774	650	431	322	(42)	
(43)		Edh at Noon	41	69	88	105	104	101	105	95	75	59	39	28	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.24	0.82	2.34	3.91	5.34	5.69	5.33	4.17	2.56	1.10	0.30	0.14	(44)	
(45)		RadStd	0.04	0.16	0.28	0.38	0.50	0.43	0.52	0.42	0.31	0.18	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A	N/A

Nomenclature: See separate page